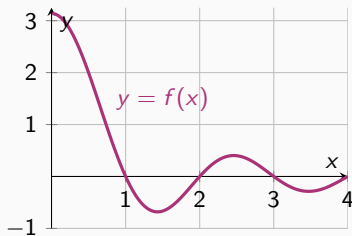


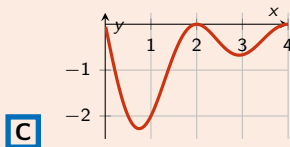
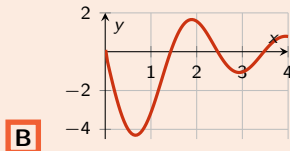
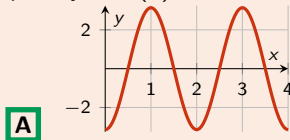
Warm up: Geometric properties of the derivative

Consider the function whose graph is shown below.



Poll Question

Which of the following shows the graph of $y = f'(x)$?



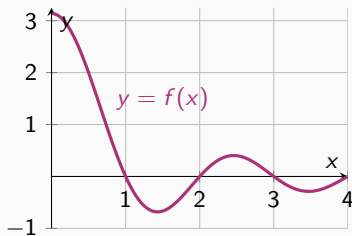
Lecture 8 Interpretations of the Derivative

Big Questions

- How are units important in interpreting derivatives?
- How can we estimate the value of a function based on limited data?
- If x changes “just a little bit”, how much will $f(x)$ change?

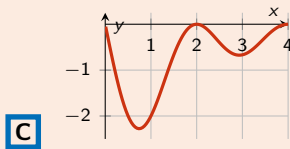
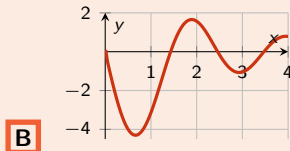
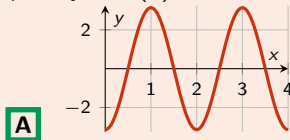
Warm up: Geometric properties of the derivative

Consider the function whose graph is shown below.



Poll Question

Which of the following shows the graph of $y = f'(x)$?



Interpretation and Units of the Derivative

Poll Question

Let C be the cost in dollars required for producing q litres of cold brew coffee. Here, $C = f(q)$.

(i) What are the **units** of $f'(2000)$?

A litres

B \$/litre

C \$/day

D litres/day

(ii) In non-technical terms, what is the meaning of $f'(2000) = 4.5$?

Leibniz Notation for the Derivative

In the last example, $C = f(q)$.

Since $f'(q) \approx \boxed{}$, we write $f'(q) = \boxed{}$.

This notation helps us remember the meaning of the derivative, as well as its units. To specify the derivative at a point in this notation, we use a vertical line:

$$f'(2000) = \boxed{}$$

Discussion Question

Let F be the amount of fuel Mr. Bean has used after driving x kilometres, t hours since he left home. Express the following using Leibniz notation:

(a) Mr. Bean's car is using fuel at a constant rate of 8 L/km.

(b) Two hours after he left home, he was traveling at a speed of 100 km/h.

Mr. Bean drives erratically, often speeding up and slowing down, but always driving down a straight road in one direction. We know that **1 hour after leaving home, he has travelled 70 km and his current speed is 36 km/h.**

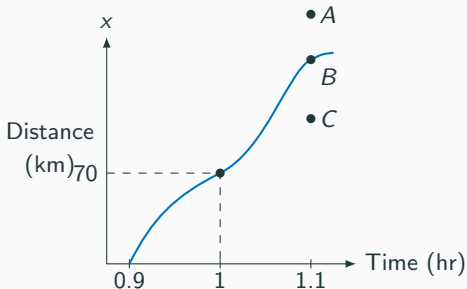


Discussion Question

- (a) About how far is he from home 1.1h after leaving?
- (b) If Mr. Bean's distance from home x (in km) as a function of time t (in h) is given by $x = f(t)$, express his speed of 36 km/h in both prime and Leibniz notation.
- (c) Summarize the calculation you did in (a) using function notation.

Illustrating Our Solution

Below, we have started a (zoomed-in) **distance-time graph** for Mr. Bean.



Poll Question

Which of the three points A , B , C could represent the estimate we made for how far from home Mr. Bean is after 1.1h? Why?

A

A

B

B

C

C

D

None are possible.

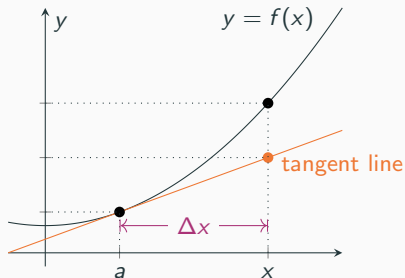
Conclusion: Our estimate lies on the

to

the graph at

.

Estimates more generally



Poll Question

Suppose we only know the values of $f(a)$ and $f'(a)$ and that x is close to a .

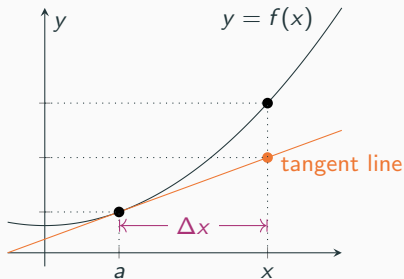
(a) Which of the following is approximately equal to $f(x) - f(a)$?

- A** $f'(a)$ **B** $f'(a)x$ **C** $f'(a)(x - a)$ **D** None

(b) Find an estimate for $f(x)$ expressed in terms of $a, x, f(a)$ and $f'(a)$.

$$f(x) \approx$$

Tangent Line Approximation: Local Linearity



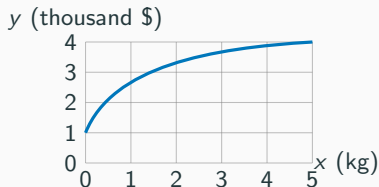
Definition

Suppose that we know $f(a)$ and $f'(a)$. As long as $\Delta x = x - a$ is **small**, to estimate $f(x)$ we can use:

This is known as the **tangent line approximation**.

Tangent Line Approximation: Local Linearity

Suppose that the cost $y = f(x)$, in thousands of dollars, of manufacturing x kilograms of ibuprofen is shown on the right.



Poll Question

- (a) It is known that $f(1) = 2.6$ and $f'(1) = 0.6$. Estimate the cost of producing 1.15 kg of ibuprofen.
- (b) Is your estimate an **A** overestimate or **B** underestimate? Why?

To do before next week:

- Please read Section 2.3.
- Finish the Week 2 Checkpoint Workbook.
- Finish Homework 2.
- The Week 2 Reflection Journal is due next Tuesday.
- The **individual part** and **pod contract** for Project 1 are due next Tuesday.

Have a great day!

Example: Rabbits

The population P of a species of rabbits as a function of time t in years since Jan 1, 2020 is shown in the table below.

Time t (years)	0	0.25	0.5	0.75	1
Population P	100	110	121	133	146

Poll Question

The rate of change $\frac{dP}{dt}$ is approximately constant, because P always grows by about 10% over each quarter year.



A True



B False

Relative Rate of Change

Definition

Let $Q = f(t)$. The **relative rate of change** of Q with respect to t , at $t = a$ is defined as

Example 1

The area of Brazil's rainforest $R = f(t)$, in million acres, is a function of the number of years t since 2000. According www.rain-tree.com, $f(9) = 740$ and $f'(9) = -2.7$. Find and interpret the relative rate of change of $f(t)$ when $t = 9$.